8 Fig. 8.1 shows two vertical, cylindrical tubes and a cylindrical magnet all held in a vacuum.

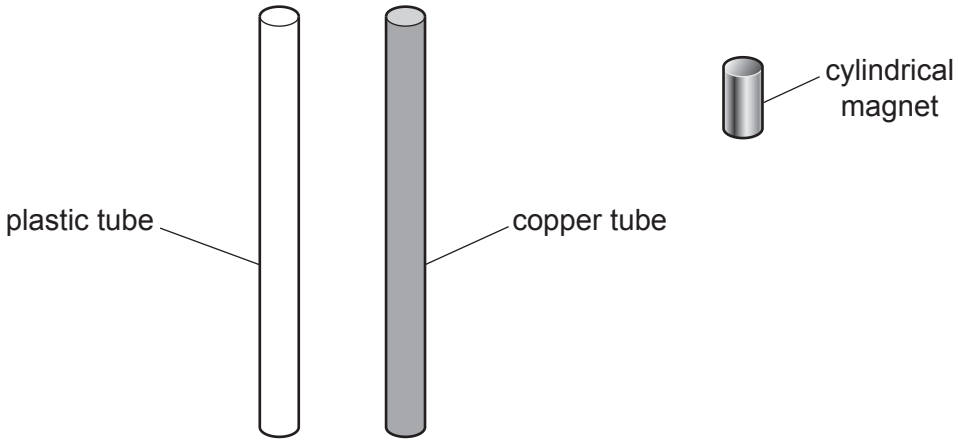


Fig. 8.1 (not to scale)

One tube is made of plastic and the other tube is made of copper. The two cylindrical tubes have identical dimensions.

The magnetic field of the small, cylindrical magnet is extremely strong.

Initially, the magnet is at rest at the top of the plastic tube.

The magnet is released and it falls through the plastic tube without experiencing a resistive force. The magnet takes 0.67 s to fall to the lower end of the plastic tube.

(a) The mass of the magnet is 0.012 kg.

Calculate the kinetic energy of the magnet when it reaches the lower end of the plastic tube.

kinetic energy = ... [4]

(b) The magnet is then held at the top of the copper tube and released. As it falls through the copper tube, an electric current is generated in the copper.

(i) Explain why there is a current in the copper.

.....

.....

..... [2]

(ii) The current in the copper produces a magnetic field of its own in the tube.

The magnet falls much more slowly in the copper tube than in the plastic tube.

Explain why the magnet falls more slowly in the copper tube.

.....

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..... [2]

[Total: 8]